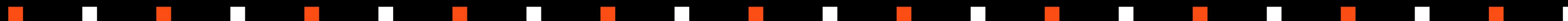


Cognitive Capital in the Age of Algorithmic Intelligence: Reimagining Human Investment for a Multipolar, AI-Augmented Economy

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INTRODUCTION



MY EXPERTISE

? +5 yr - Public Policy and Interculturality
Researcher: MSc for understanding
whys and *wherefores*.

⇒ + 5 yr - Innovation Ecosystems
Researcher: BSc Math & Law for
proposing *whats*.

\$ +20 yr - Technologist: BSc Eng & MSc
for building *hows* thru **strategic use** of
Technology.

#Cognitive Capital

#Multipolar Economic Development

#AI-augmented Economy

#AI-driven Labour

global growth new platform = f(AI-enhanced Human Capital investments, Labour)

MY IDEA

Machines becoming **smarter than us**: how do we make sure that
people don't get left behind?

Build '**Cognitive Capital**' —human
ability to grow and evolve
alongside AI. **CN¥125 trillion +GDP**

Multipolar nations turning AI
revolution into a human evolution
through smarter state investments.

WHY?

NOW is the time of imperativeness:

Tech **sovereignty** determines **civilisational trajectories**,
not merely military nor economic power.

1. Crisis on the rise: AI-Labour gap widening

- **85 million jobs displaced** by 2025 (WEF, 2020).
- Tech change velocity –**exponential**– vs institutional response capability –linear– (Azhar, 2021).
- Industrial model obsolete: **professionals for 2010** (static skills, 4-year degree cycles, reactive reskilling).
- **23% youth unemployment** rates by 2030 (OECD, 2023).

2. The opportunity: need **NEW** Human Capital Investments

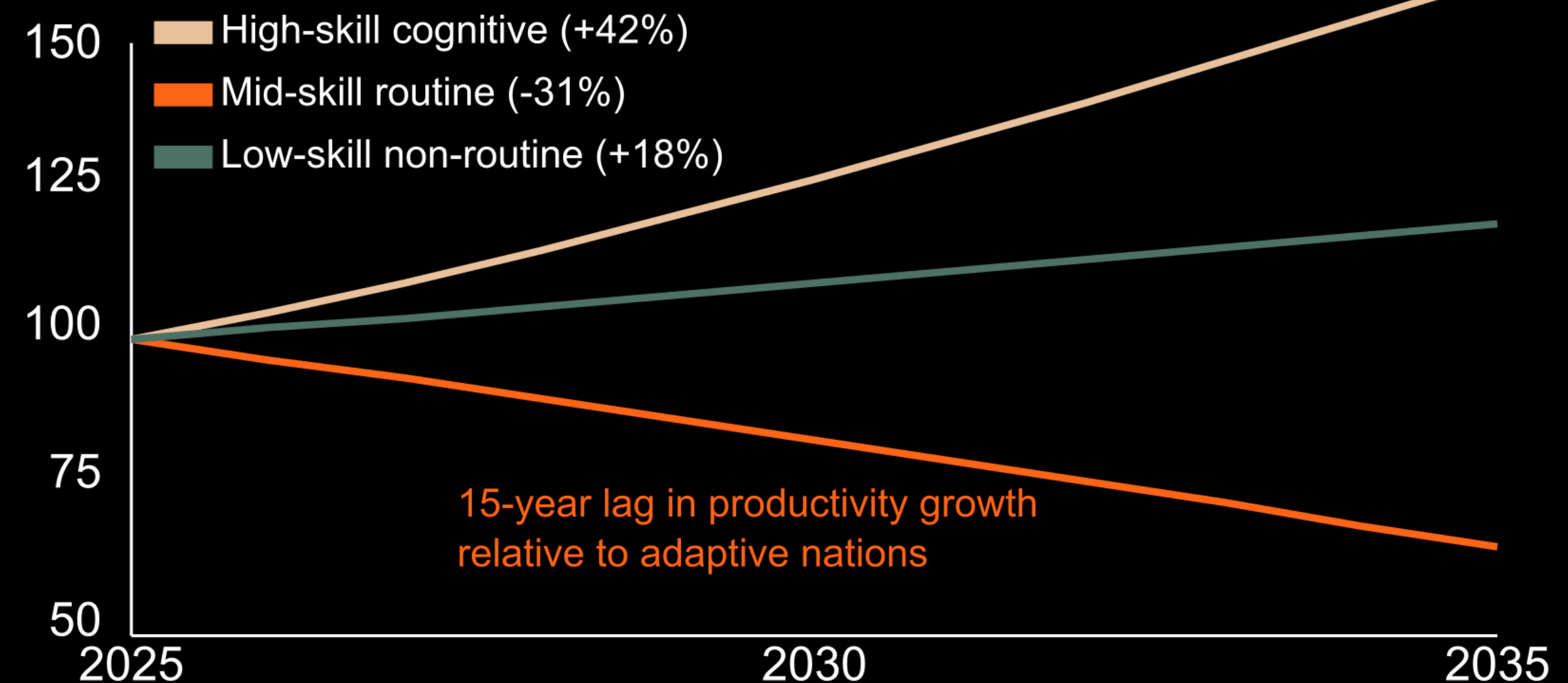
- 97 million new roles by 2025 (WEF, 2020).
- 375 million workers switching due to AI (McKinsey, 2021).
- BlackSwan emergence: multipolar **civilisational clusters**.

2. The stakes

- Without action → 32% of AI gains captured by top 1%.
- With investment → 2.3x faster recovery, **34% unemployment reduction**

Labour Market Polarisation Dynamics

'Missing middle' crisis: backbone of societies dissapears while extremes expand



Source: own elaboration based on datasources from Autor & Dorn (2023) and Frey & Osborne (2017).

WHAT?



Cognitive Capital cultivation

A dynamic **lifelong** process of Human-AI **coevolution** structured across three interdependent horizons:

1. Horizon One (2025- 2030): tactical augmentation and displacement management

- Tactical reskilling; 6-12 month programs; AI-adjacent fields.
- Displacement management; UBI pilots; Transition funds; Micro-credentials.

2. Horizon Two (2030-2040): **strategic** capability redesign

- Capability redesign; Human-AI complement; Mission-oriented frameworks.
- Multipolar models; Collective intelligence; *Leapfrog* innovation.

3. Horizon Three (2040-2050): **Civilisational Learning Systems**

- Learning civilisations; *Liquid universities*; Lifelong accounts; Adaptive Intelligence Quotient metric.
- Multipolar cooperation; 'Geneva Convention' for AI-Labour; Credential portability.

Divergent Civilisational Trajectories

Predictive scenarios for Russian Federation-China-Global South (2025-2035)

Trajectory Alpha China adaptive autarky	Trajectory Beta Russia sovereign fortress	Trajectory Gamma Global South leapfrog
Probability: 68% Productivity: 6.2% annual growth	Probability: 54% Productivity: 3.8% annual growth	Probability: 45% Productivity: 7.1% annual growth
85% workforce in continuous learning by 2035	40% workforce in digital sovereignty sectors by 2035	Demographic dividend + institutional flexibility
Trade-off: Reduced autonomy, State career pathways, innovation ceiling risk	Trade-off: Reduced global access, strategic concentration, regional inequality	Risk: Uneven development, intra-national inequality, potential instability

Source: own elaboration based on datasources from Autor & Dorn (2023), Frey & Osborne (2017), OECD (2023), McKinsey (2021), Acemoglu & Restrepo (2020).

HOW?



Five actionable imperatives for Human Capital Investments

Investing in the ultimate renewable resource: *Cognitive Capital*, continuously cultivated, strategically deployed, and ethically governed.

National Cognitive Capital Agencies

- Authority equivalent to central banks
- Coordinate lifelong learning
- Credential standards
- Human-AI integration across sectors

Policy

Dynamic Credentialing Infrastructure

- Replace degrees with continuous competency validation
- Blockchain-verified & AI-updated
- Real-time capability portfolios

Infrastructure

AI Productivity Dividend Systems

- 2-3% levy on algorithmic gains for funding universal learning accounts & Transition support
- Intercultural innovation: *Liquid Universities* permeating across blocs

Financing

Multipolar Learning Standards

- Credential portability & cross-border talent mobility within BRICS+
- Ethical AI in education
- "Geneva Convention for AI-Era Labor"

Coordination

Human Capital Vitality Metrics

- Track adaptive capacity
- Skill acquisition speed
- Economic mobility across career transitions —not just employment rates

Measurement

Cognitive Capital Ecosystem



Economics: (productivity)^{ai}

- Productivity paradox resolution
- 5.7x gains with human upskilling
- 1.8x higher rates in entrepreneurship
- 67% of the multipolar bloc workforce engaged in continuous learning
- 43% reduction in structural unemployment by displacement
- **CN¥125-165 trillion additional GDP** Russian Federation-China-Global South by 2035



A new Social Dimension

- Identity crisis: work redefined
- 5-7 career transformations/lifetime
- Demographics+: leapfrog in youth, elders transforming collective data into civilisational wisdom
- Governance legitimacy threshold: democratic stability risk overcome
- Emergence of a new social contract: State guarantees learning opportunity, individuals commit to adaptive capability civilisational development



Inequality Dynamics

- No intervention: top 1% captures 32% of AI gains while bottom 50% net income decline
- Labour polarisation: High-skill: +42%; Middle-skill: -31%; Low-skill: +18%
- 23% youth unemployment rates by 2030
- 15-year lag in productivity growth relative to adaptive nations
- Accelerating brain drain (8-12% annual emigration of high-skilled workers).

CONCLUSION: 5 MT



Not responding to megatrends...
Designing systems that harness them for **human flourishing**

5 Megatrends		<i>Cognitive Capital</i> addresses:
1. Fragmented Globalization	→	• Multipolar coordination
2. Platformization of Economy	→	• Sovereign Human-AI symbiosis
3. Financial Transformation	→	• Productivity dividends
4. Demographic Catastrophe	→	• Adaptive learning systems
5. Human Capital Era	→	• Continuous capability cultivation

Civilisations that thrive will not be those with the most advanced AI, but those that most effectively invest in the **ultimate renewable resource**:
HUMAN COGNITIVE CAPITAL
continuously cultivated, strategically deployed, ethically governed...

THE CHOICE IS OURS: We don't ask IF humans stay relevant...
We DESIGN the systems that determine relevance itself.